Bihar Engineering University, Patna

B.Tech 5th Semester Examination, 2025

Course: B.Tech | ***Time: 03 Hours***
Code: 101503 | ***Subject: Geotechnical Engineering–I*** | ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
*(iv) Question No. **1** is compulsory.*

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) Soil mechanics mainly deals with the study of
(i) Steel structures (iii) Behaviour of soils under loads
(ii) Fluid flow only (iv) Rock blasting

(b) Void ratio is defined as
(i) Volume of solids/volume of voids (iii) Volume of voids/volume of solids
(ii) Total volume/volume of solids (iv) Volume of water/volume of air

(c) The water content of soil is commonly determined by
(i) Pycnometer method (iii) Core cutter method
(ii) Oven drying method (iv) Sand replacement method

(d) Liquid limit of soil is determined using
(i) Proctor apparatus (iii) Triaxial apparatus
(ii) Direct shear apparatus (iv) Casagrande apparatus

(e) Darcy's law is valid for
(i) Turbulent flow (iii) Unsteady flow
(ii) Compressible flow (iv) Laminar flow

(f) The coefficient of permeability is expressed in
(i) m/s (iii) kg/m³
(ii) m²/s (iv) N/m²

(g) Effective stress principle was proposed by
(i) Rankine (iii) Terzaghi
(ii) Boussinesq (iv) Coulomb

(h) Quick sand condition occurs when
(i) Effective stress becomes zero (iii) Water content becomes zero
(ii) Soil becomes dry (iv) Unit weight increases

(i) Optimum moisture content is obtained from
(i) Consolidation test (iii) Shear test
(ii) Permeability test (iv) Compaction test

(j) Newmark's influence chart is used for
(i) Soil classification (iii) Stress distribution analysis
(ii) Flow net construction (iv) Compaction control

Q.2 (a) Explain the formation and classification of soils with suitable examples. **[7]**
(b) Discuss the three-phase system of soil and derive the important phase relationships. **[7]**

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Q.3	(a)	Explain various laboratory methods used for determination of moisture content in soils.	[7]
	(b)	Describe the methods for determination of specific gravity and field density of soils	[7]
Q.4	(a)	Explain Atterberg limits and discuss their engineering significance.	[7]
	(b)	Discuss the Indian Standard Soil Classification System with neat illustrations.	[7]
Q.5	(a)	Explain Darcy's law and discuss the factors affecting permeability of soils.	[7]
	(b)	Describe constant head and falling head permeability tests with suitable sketches.	[7]
Q.6	(a)	Explain capillary action and seepage pressure in soils with engineering applications.	[7]
	(b)	Describe laboratory and field compaction methods used in geotechnical engineering.	[7]
Q.7	(a)	Explain the factors affecting compaction and discuss Proctor compaction test.	[7]
	(b)	Discuss stresses in soils due to point load and line load using Boussinesq's theory.	[7]
Q.8	(a)	Explain the stress distribution under uniformly loaded circular and rectangular areas.	[7]
	(b)	Discuss Newmark's influence chart and its applications in stress analysis of soils.	[7]

Q.9 Write short note on ***any two*** of the following: **[7 x 2 = 14]**

(a) Quick Sand Condition
(b) Liquidity Index and Consistency Index
(c) Flow Net Characteristics
(d) Compaction Control in Field

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